

Theory of Computation

Regular Languages and Regular Expressions

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Reverse of a Regular Language

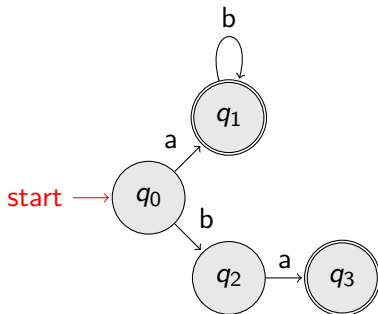
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Reverse of a Regular Language

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Construct NFA that accepts L^R : invert the transitions of the NFA that accepts L .

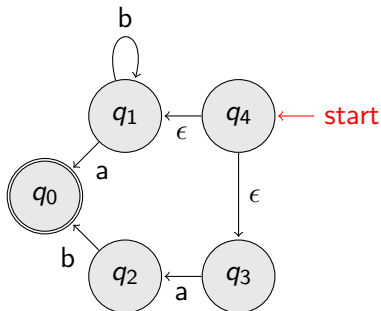
Reverse of a Regular Language

- **Theorem:** The reverse L^R of a regular language, L is a regular language
- **Proof idea:**
Construct NFA that accepts L^R : invert the transitions of the NFA that accepts L .
- **Proof:** Since L is regular, there is NFA that accepts L .
Example: $L = ab^* + ba$



Reverse of a Regular Language

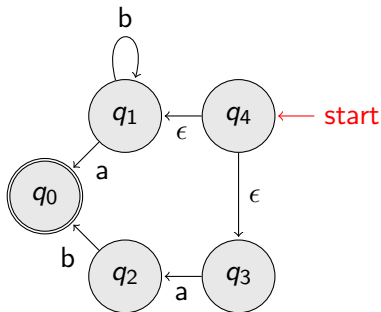
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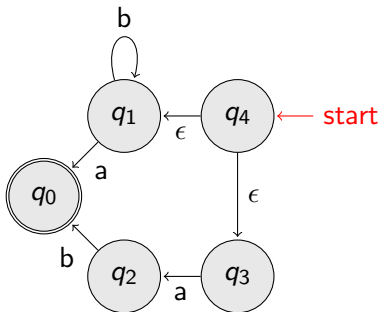
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Reverse of a Regular Language

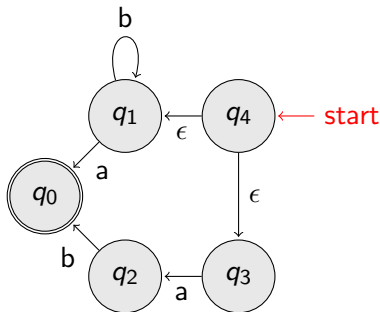
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Reverse of a Regular Language

- **Proof:** Invert Transitions.



- Make old initial state a final state
- Add a new initial state
- $L^R = b^*a + ab$
- Resulting machine accepts L^R ; So L^R is regular.

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- Grammars express languages.

- Example: the English language**

$\langle \text{Sentence} \rangle \rightarrow \langle \text{Noun_Phrase} \rangle \langle \text{Predicate} \rangle$

$\langle \text{Noun_Phrase} \rangle \rightarrow \langle \text{Article} \rangle \langle \text{Noun} \rangle$

$\langle \text{Predicate} \rangle \rightarrow \langle \text{Verb} \rangle$

$\langle \text{Article} \rangle \rightarrow a | the$

$\langle \text{Noun} \rangle \rightarrow boy | dog$

$\langle \text{Verb} \rangle \rightarrow runs | walks$

- A derivation of “the boy walks”:
 $\langle \textit{Sentence} \rangle \implies \langle \textit{Noun_Phrase} \rangle \langle \textit{Predicate} \rangle$

- A derivation of “the boy walks”:

$$\begin{aligned} \langle \textit{Sentence} \rangle &\implies \langle \textit{Noun_Phrase} \rangle \langle \textit{Predicate} \rangle \\ &\implies \langle \textit{Noun_Phrase} \rangle \langle \textit{Verb} \rangle \end{aligned}$$

- A derivation of “the boy walks”:

$\langle \text{Sentence} \rangle \implies \langle \text{Noun_Phrase} \rangle \langle \text{Predicate} \rangle$
 $\implies \langle \text{Noun_Phrase} \rangle \langle \text{Verb} \rangle$
 $\implies \langle \text{Article} \rangle \langle \text{Noun} \rangle \langle \text{Verb} \rangle$

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$\langle \text{Sentence} \rangle \implies \langle \text{Noun_Phrase} \rangle \langle \text{Predicate} \rangle$
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 $\implies \langle \text{Article} \rangle \langle \text{Noun} \rangle \langle \text{Verb} \rangle$
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- A derivation of “the boy walks”:

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 $\implies \text{the} \langle \text{Noun} \rangle \langle \text{Verb} \rangle$
 $\implies \text{the boy} \langle \text{Verb} \rangle$

- A derivation of “the boy walks”:

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 $\implies \text{the boy walks}$

References

- Denial I. A. Cohen Introduction to Computer Theory, Second Edition, John Wiley & Sons, 1997.
- J. E. Hopcroft, R. Motwani, & J. D. Ullman Introduction to Automata Theory, Languages, and Computation, Third Edition, Pearson, 2008.